Project Report

On

HR ANALYTICS IN POWER BI

Course: AI for Managers: Application and Strategy

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Dataset Link: https://www.kaggle.com/datasets/hopesb/hr-analytics-dataset

Executive Summary

This project analyses the HR dataset, covering 2,845 employee records across 32 fields — 28 source columns plus 4 derived analytical fields. Every figure in this report was computed directly from the dataset and its associated Power BI model; none is estimated or assumed.

Overall attrition stands at 13.60% (387 of 2,845 employees). The workforce is concentrated in Production (1,910 employees, 67.1%) and in Massachusetts (2,523 employees, 88.7%). Software Engineering carries the highest departmental attrition rate (17.86%), narrowly ahead of Production (16.86%). Engagement, satisfaction, and work-life balance scores each average close to the midpoint of their 1–5 scales. Attrition is highest among Low-Engagement employees (14.04%) and, counter-intuitively, among High-Satisfaction employees (15.04%) — satisfaction score alone does not reliably predict attrition here. Employees on a Performance Improvement Plan show the highest attrition of any performance band (20.93%).

Training data shows Internal programmes slightly outperforming External on success rate (52.50% vs 49.16%) on near-identical spend. Communication Skills is the strongest programme (54.34% success); Technical Skills the weakest (45.12%). Recommendations center on the largest measurable risk segment: Production employees with Low Engagement (833 employees, 17.41% attrition).

Business Problem and Objectives

Organizations hold rich employee data but often lack a consolidated way to analyse it, leaving basic questions — where attrition concentrates, which segments are disengaged, whether training spend works — unanswered. A well-structured dataset with a proper model and verified measures allows these questions to be answered consistently and repeatably.

Objectives

- Workforce analysis: profile headcount by department, division, geography, and demographics.
- Attrition analysis: quantify attrition overall and by segment.
- Diversity analysis: examine gender, race, marital status, and age distribution.
- Engagement and performance analysis: assess engagement, satisfaction, work-life balance, and performance, and their link to attrition.
- Training effectiveness: evaluate training cost, success rate, and outcomes.
- HR recommendations: translate findings into concrete, evidence-based actions.

Dataset Overview

Dataset: 2,845 rows, 32 columns (28 source + 4 derived), one row per employee, keyed on Employee ID.

Key field groups: identity (Employee ID, EmployeeStatus, DepartmentType, Division, Title), geography (State), dates (StartDate, DOB, Survey Date, TrainingDate), engagement/performance (Engagement, Satisfaction, Work-Life Balance, Performance Score, Current Employee Rating), and training (Programme, Type, Outcome, Duration, Cost).

Data Dictionary

Field	Type	Description
Employee ID	Int64	Unique employee identifier (0 duplicates)
StartDate	Date	Employment start date
Title / BusinessUnit / DepartmentType / Division	Text	Role and organizational placement
EmployeeStatus	Text	Active or Terminated

Field	Type	Description
EmployeeType / PayZone / EmployeeClassificationType	Text	Employment and pay classification
DOB	Date	Date of birth
State	Text (Geography)	US state
GenderCode / RaceDesc / MaritalDesc	Text	Demographic fields
Performance Score / Current Employee Rating	Text / Int64	Performance rating (label and numeric)
Survey Date / Engagement Score / Satisfaction Score / Work-Life Balance Score	Date / Int64	Engagement survey fields, 1–5 scale
TrainingDate / Training Program Name / Training Type / Training Outcome	Date / Text	Training record fields
TrainingDurationDays / Training Cost	Int64 / Double	Training duration and cost
Age	Int64	Employee age
Age Group (calculated)	Text	Under 25 / 25–34 / 35–44 / 45–54 / 55+
Engagement / Satisfaction Category (calculated)	Text	Low / Medium / High
Training Success (calculated)	Text	Successful / Unsuccessful

Data Cleaning and Preprocessing

Steps below reflect the actual Power Query (M) transformation applied to the dataset.

Power Query Steps

- Source: Excel workbook containing the HR employee data.
- Promoted first row as headers.
- Explicit data types assigned to all 28 source columns.
- Removed fully blank rows.
- Employee ID trimmed of whitespace, then reconverted to whole number.
- Renamed “Training Date” → “TrainingDate” and “Training Duration(Days)” → “TrainingDurationDays” for consistency.

Derived Fields

Four analytical fields were added at the model layer: Age Group, Engagement Category, and Satisfaction Category (each bucketed Low/Medium/High or by age band from the underlying score), and Training Success (Successful where Training Outcome is Completed or Passed; Unsuccessful where Failed or Incomplete).

Data-Quality Results

Missing values: 0 across all 32 columns. Duplicate Employee IDs: 0. Clean data types and a verified unique key are what make every measure in this report reliable.

Data Model, DAX, and KPI Framework

A calculated Date table spans 1,826 days (2018-08-07 to 2023-08-06), covering StartDate, Survey Date, and TrainingDate combined, and is marked as an official Date table.

StartDate carries the active relationship to the Date table, since core headcount and hiring visuals depend on it. Survey Date and TrainingDate are linked but inactive, activated inside specific measures via USERELATIONSHIP so engagement- and training-dated trends resolve correctly without conflicting with the primary hiring timeline.

Key Measures

Measure	Logic	Definition
Total Employees	COUNTROWS(…)	Count of all employee records.
Active / Terminated Employees	CALCULATE(COUNTROWS(…), EmployeeStatus=…)	Count filtered to each status.
Attrition Rate	DIVIDE(Terminated, Total, 0)	Share of employees Terminated.
Average Age / Engagement / Satisfaction / Work-Life Balance / Employee Rating	AVERAGE(…)	Mean of each respective field.
Total / Average Training Cost	SUM(…) / AVERAGE(…)	Aggregate and mean training spend.
Training Success Rate	DIVIDE(Successful count, Total, 0)	Share of training records marked Successful.
Total Training Cost (by Training Date)	CALCULATE(…, USERELATIONSHIP(…))	Training cost trended by the date training occurred.

Dashboard Design and Implementation

The dashboard is organized into five pages: Executive Overview (headline KPIs and headcount trend), Workforce and Diversity (department, division, age, gender, pay zone breakdowns), Attrition Analysis (attrition by department, state, age, engagement, and satisfaction), Engagement, Performance, and Training (score distributions and training outcomes), and AI Insights and Recommendations (Key Influencers, Decomposition Tree, Smart Narrative, and written recommendations).

Card visuals communicate single headline KPIs; bar and column charts compare categorical fields (department, pay zone, age group); a map visual uses the State field's Geography data category; line charts against the Date table show trends; and matrix/table visuals present exact figures where precision matters more than visual estimation.

Executive Overview Findings

Total employees: 2,845. Active: 2,458 (86.4%). Terminated: 387 (13.6%). Overall attrition rate: 13.60%.

Hiring Trend (New Hires by Start Year)

2023 is a partial year — the dataset's latest date is 2023-08-06.

Year	New Hires
2018	248
2019	560
2020	560
2021	564
2022	594
2023 (partial)	319

Department Size

Department	Employees	% of Workforce
Production	1,910	67.1%
IT/IS	409	14.4%
Sales	311	10.9%
Software Engineering	112	3.9%
Admin Offices	79	2.8%
Executive Office	24	0.8%

Massachusetts holds the largest employee population at 2,523 (88.7%) across 28 states; no other state exceeds 35 employees, indicating a strongly centralized workforce.

Workforce and Diversity Findings

Age Group

Age Group	Employees	% of Workforce
55+	1,213	42.6%
25–34	493	17.3%
35–44	459	16.1%
45–54	429	15.1%
Under 25	251	8.8%

Gender

Female: 1,588 (55.8%). Male: 1,257 (44.2%). Production and IT/IS skew female; Sales and Executive Office skew male.

Department	Female	Male
Production	1,190	720
IT/IS	199	210
Sales	109	202
Software Engineering	46	66
Admin Offices	43	36
Executive Office	1	23

Race and Marital Status

Largest race group: Asian, 598 (21.0%), narrowly ahead of Black (579, 20.4%); the remaining groups (White 572, Other 556, Hispanic 540) fall within a similarly narrow band — no group dominates. Largest marital-status group: Single, 731 (25.7%), narrowly ahead of Married (730); all four categories sit within a 47-employee range, indicating a near-even split.

Average Age by Department

Average age is tightly clustered between 48.4 and 49.8 years across all six departments, so age does not meaningfully differentiate departments in this dataset.

Attrition Analysis Findings

Every figure below is an association between a segment and attrition rate; none establishes causation.

By Department

Department	Employees	Attrition Rate
Software Engineering	112	17.86%
Production	1,910	16.86%
IT/IS	409	8.31%
Sales	311	3.22%
Admin Offices	79	1.27%

By State (Highest, Plus Largest State)

State	Employees	Attrition Rate
PA	13	38.46%
TX	35	25.71%
MA (largest state)	2,523	14.70%
CT	33	3.03%

By Age Group

Age Group	Employees	Attrition Rate
35–44	459	14.60%
45–54	429	14.45%
25–34	493	13.39%
Under 25	251	13.55%
55+	1,213	13.03%

By Pay Zone

Pay Zone	Employees	Attrition Rate
Zone B	934	14.35%
Zone A	1,013	14.31%
Zone C	898	12.03%

By Engagement / Satisfaction / Work-Life Balance

Attrition is highest among Low-Engagement employees (14.04%), but — counter-intuitively — also highest among High-Satisfaction employees (15.04%), and follows the expected pattern for Work-Life Balance (Low 14.60% vs High 12.90%).

Engagement	Employees	Attrition Rate
Low	1,204	14.04%
Medium	532	13.53%
High	1,109	13.17%
Satisfaction	**Employees**	**Attrition Rate**
High	1,177	15.04%
Low	1,099	13.19%

Engagement	Employees	Attrition Rate
Medium	569	11.42%
Work-Life Balance	**Employees**	**Attrition Rate**
Low (≤2)	1,130	14.60%
Medium (=3)	599	13.02%
High (≥4)	1,116	12.90%

By Performance Score

PIP-rated employees show the highest attrition of any performance band (20.93%), consistent with an expected relationship between low formal standing and separation.

Performance Score	Employees	Attrition Rate
PIP	86	20.93%
Fully Meets	2,251	13.73%
Needs Improvement	162	12.35%
Exceeds	346	11.56%

Engagement and Performance Findings

Company-wide averages: Engagement 2.94, Satisfaction 3.03, Work-Life Balance 2.99 (1–5 scale). Among departments with a reliable base, IT/IS leads engagement, Sales leads satisfaction (3.14, the company-wide high), and Admin Offices leads work-life balance.

Department	Avg Engagement	Avg Satisfaction	Avg Work-Life Balance
IT/IS	3.02	3.01	2.98
Software Engineering	2.96	3.10	2.99
Sales	2.98	3.14	3.03
Admin Offices	2.95	2.51	3.20
Production	2.91	3.03	2.97

Company-wide average employee rating: 2.97. The largest division, Field Operations (747 employees), rates above average at 3.05.

Performance-Score Distribution

Roughly 4 in 5 employees (79.1%) fall into the “Fully Meets” band.

Performance Score	Employees	% of Workforce
Fully Meets	2,251	79.1%
Exceeds	346	12.2%
Needs Improvement	162	5.7%
PIP	86	3.0%

Terminated vs. Active Employees

Terminated employees have a higher average performance rating than Active employees, with engagement close to but slightly below Active — a pattern consistent with “regretted attrition” (the loss of employees who were not underperforming), though the dataset alone cannot confirm the cause.

Metric	Active	Terminated
Average Employee Rating	2.96	3.07
Average Engagement	2.95	2.90

Training Effectiveness Findings

Internal training shows a higher success rate than External (52.50% vs 49.16%) on near-identical spend and participation.

Type	Employees	Total Cost	Success Rate
Internal	1,421	$797,257.16	52.50%
External	1,424	$793,891.47	49.16%

Training Programme Performance

Communication Skills is the strongest programme (54.34% success, also the highest spend). Technical Skills is the weakest (45.12%) despite being the lowest-cost programme. Project Management is the clearest high-cost/low-success combination: second-highest spend at a below-average 47.69% success rate.

Programme	Employees	Total Cost	Success Rate
Communication Skills	633	$344,544.00	54.34%
Customer Service	540	$307,736.53	53.52%
Leadership Development	544	$305,414.47	53.13%
Project Management	585	$329,309.80	47.69%
Technical Skills	543	$304,143.83	45.12%

Training Outcome

Outcomes are fairly evenly distributed, with Completed (25.9%) the largest and Failed (23.5%) the smallest — a 2.4-point spread indicating no single outcome dominates.

Outcome	Employees	% of Records
Completed	737	25.9%
Incomplete	731	25.7%
Passed	709	24.9%
Failed	668	23.5%

AI Features Used

This section explains how each feature applies to this model; it does not assert a specific finding was produced live unless independently confirmed in Power BI Desktop.

Key Influencers

Pointed at EmployeeStatus = “Terminated” as the target, using DepartmentType, Engagement Category, PayZone, Age Group, and Performance Score as candidate influencers, Key Influencers ranks which fields are most associated with attrition. The directional patterns already established via direct analysis — Software Engineering and Production carrying the highest departmental attrition, Low Engagement carrying the highest engagement-linked attrition — are the kind of relationship this visual is designed to surface.

Decomposition Tree

Rooted on Terminated Employees and split by DepartmentType, State, Age Group, and Engagement Category, a Decomposition Tree lets a viewer drill from the overall 387 terminated employees into specific segments, such as Production employees with Low Engagement (833 employees, 17.41% attrition) — mirroring the cross-tabulation in this report as an interactive tool.

Smart Narrative

A Smart Narrative visual can auto-generate natural-language summaries of key figures (e.g., “Attrition Rate is 13.60%…”) that update as filters change, provided it is pointed only at measures already verified in the model.

Ethical Note

These AI-assisted findings describe statistical associations within this specific, historical dataset. They do not constitute proof of causation and should not be the sole basis for decisions about individual employees without further human review.

Answers to Questions 1–30

Questions 1–15

Q#	Question	Answer	Visual	Interpretation
1	Total employees	2,845	Executive Overview (Card)	2,845 verified employee records.
2	Active employees	2,458 (86.4%)	Executive Overview (Card)	The large majority hold Active status.
3	Terminated employees	387 (13.6%)	Executive Overview (Card)	387 employees carry Terminated status.
4	Overall attrition rate	13.60%	Executive Overview (Card)	Roughly 1 in 7 employees has left.
5	Hiring trend	2018: 248; 2019–2022: 560–594; 2023: 319 (partial)	Executive Overview (Line)	Hiring was broadly stable 2019–2022; 2023 is a partial year.
6	Largest department	Production — 1,910 (67.1%)	Workforce & Diversity (Bar)	Production employs two-thirds of the workforce.
7	Largest division	Field Operations — 747	Workforce & Diversity (Bar)	The single largest division, ahead of General - Con (474).
8	State with most employees	MA — 2,523 (88.7%)	Workforce & Diversity (Map)	Workforce is heavily concentrated in Massachusetts.
9	Age-group distribution	55+: 1,213 (42.6%); 25–34: 493; 35–44: 459; 45–54: 429; Under 25: 251	Workforce & Diversity (Bar)	55+ is the largest single age segment.
10	Gender by department	Production F1,190/M720; Sales F109/M202; IT/IS F199/M210	Workforce & Diversity (Clustered Bar)	Production and IT/IS skew female; Sales skews male.
11	Largest race group	Asian — 598 (21.0%)	Workforce & Diversity (Donut)	Narrowly the largest among an evenly distributed set.
12	Largest marital-status group	Single — 731 (25.7%)	Workforce & Diversity (Donut)	Nearly evenly split across four categories.
13	Department with highest attrition	Software Engineering — 17.86%	Attrition Analysis (Bar)	Narrowly ahead of Production (16.86%).
14	State with highest attrition	MA (largest state) — 14.70%	Attrition Analysis (Map)	The most statistically reliable reading given its scale.
15	Age group with highest attrition	35–44 — 14.60%	Attrition Analysis (Bar)	Spread across age groups is narrow (13.0–14.6%).

Questions 16–30

Q#	Question	Answer	Visual	Interpretation
16	Pay zone with highest attrition	Zone B — 14.35%	Attrition Analysis (Bar)	Zone A (14.31%) is close behind; both above Zone C (12.03%).
17	Engagement group with highest attrition	Low — 14.04%	Attrition Analysis (Bar)	Highest of the three, though the gap is modest.
18	Satisfaction group with highest attrition	High — 15.04%	Attrition Analysis (Bar)	Counter-intuitively the highest of the three bands.
19	Highest-risk employee segment	By rate: Software Engineering + Medium Engagement (21.88%, 32 employees). By volume: Production + Low Engagement (17.41%, 833 employees)	AI Insights (Decomposition Tree)	The volume segment is the largest absolute risk pool.
20	Department with highest engagement	IT/IS — 3.02 (largest reliable base)	Engagement/Performance (Bar)	Leads among departments with meaningful headcount.

Q#	Question	Answer	Visual	Interpretation
21	Department with highest satisfaction	Sales — 3.14	Engagement/Performance (Bar)	The company-wide high, on a reasonably sized base.
22	Division with highest employee rating	Field Operations — 3.05 (largest reliable)	Engagement/Performance (Bar)	Above the company average (2.97).
23	Performance-score distribution	Fully Meets 79.1%; Exceeds 12.2%; Needs Improvement 5.7%; PIP 3.0%	Engagement/Performance (Donut)	4 in 5 employees fall into Fully Meets.
24	Department with best work-life balance	Admin Offices — 3.20 (largest reliable)	Engagement/Performance (Bar)	Leads among departments with meaningful headcount.
25	Training type with highest total cost	Internal — $797,257.16	Training (Bar)	Marginally above External ($793,891.47).
26	Training type with weakest outcomes	External — 49.16% success	Training (Bar)	Underperforms Internal (52.50%) on similar spend.
27	Training programme with highest success rate	Communication Skills — 54.34%	Training (Bar)	The strongest-performing programme; Technical Skills lags at 45.12%.
28	Factors most associated with termination	Department (17.9% to 1.3% spread); PIP performance (20.93%); Low Engagement (14.04%)	AI Insights (Key Influencers)	Department and formal performance standing are the strongest observed differentiators.
29	Employee segment requiring immediate attention	Production + Low Engagement — 833 employees, 17.41% attrition	AI Insights (Decomposition Tree)	Largest absolute at-risk population.
30	Top HR priority	Engagement-driven attrition within Production	AI Insights & Recommendations	Offers the largest measurable reduction in total attrition of any single intervention identified.

Recommendations

- Production employees with Low Engagement show 17.41% attrition across 833 people, the largest risk pool by volume — launch a structured engagement intervention (manager check-ins, pulse follow-up) targeted at this segment first.
- Software Engineering has the highest departmental attrition rate (17.86%) — run stay interviews to identify department-specific retention drivers.
- Terminated employees have a higher average rating (3.07) than Active employees (2.96) with comparable engagement — conduct exit interviews focused on higher-rated leavers to investigate possible regretted attrition.
- External training underperforms Internal (49.16% vs 52.50%) and Project Management is the clearest high-cost/low-success programme ($329,309.80 at 47.69%) — audit External vendors and Project Management content, and reallocate budget toward Communication Skills.
- Attrition in Pay Zones A and B (14.31–14.35%) runs above Zone C (12.03%) — conduct a compensation/benefits equity review; this is a correlation worth investigating, not a confirmed cause.
- The High-Satisfaction group shows the highest attrition of any satisfaction band (15.04%) — investigate directly (exit interviews or a follow-up survey) before treating satisfaction score as a reliable retention indicator here.

Project Limitations

- No exit-date or exit-reason field is present; attrition can only be measured as a status, not trended by when or why an employee left.
- No compensation/salary field is present; Pay Zone allows only a categorical, not a precise monetary, view.
- No recruitment-source, promotion-history, or absence/overtime field is present.
- Geographic analysis is constrained by extreme concentration in one state (88.7% in MA); other-state findings rest on very small samples.
- Key Influencers, Decomposition Tree, and Smart Narrative outputs described in this report reflect how these features function methodologically; their live output in the current dashboard was not independently re-verified for this document.

Future Analysis Opportunities

- Attrition prediction: a classification model using the verified engagement, satisfaction, performance, and demographic fields.
- Salary/compensation equity analysis, if a salary field is obtained.
- Overtime and absence analysis, if attendance data is obtained.
- Promotion and career-progression analysis, if promotion-history data is obtained.
- Recruitment-source analysis, if hiring-channel data is obtained.
- Training ROI analysis, extending cost/success-rate data with post-training performance change.
- Exit-reason analysis, if exit-interview data is obtained, to move toward causal explanations.